Theory of Everything (ToE) — Strict Final Documentation

Release 7 (Strict) — Full Definitions, Derivations, and Discharge Map

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Abstract

This article is the self-contained *Release 7 (Strict)* documentation target of the ToE program in this repository. It records (i) the strict core candidate Lagrangian template (one-kernel nadsoliton core), (ii) the strict global selector closure achieved in *projective (ray-level)* scope on the declared configuration space C_{v1}, and (iii) the resulting operational computability pipeline ("ToE OS") producing first strict quadratic physics proxies and observer-limit downstream artifacts. All stronger claims (kernel-alone/global discharge of the strict uniqueness obstruction `QW-2191`, any directed/sign-sensitive physical orientation datum in strict core, Standard Model identification/host matching in strict scope, and strong ToE closure) remain explicitly separated and are **not** silently promoted.

Reproducibility is provided by the repository's exported JSON artifacts and smoke probes; see the *Reproducibility stamp* below.

Origin and philosophy (required fragment)

The theory originates from a deep intuition that **Information is the fundamental substance of reality**, consistent with the metaphysical insight that *"In the beginning was the Word"* (Logos/Information). This intuition evolved through key realizations:

1. **Eucharistic Inspiration:** A profound fascination with the memorial of the **Eucharist of Jesus Christ** and its material manifestation in reality served as the primary inspiration, suggesting a direct mechanism by which spiritual/informational reality can condense into tangible matter.
2. **Fractal Nature:** Observing self-similarity across vast scales suggested that fundamental information must possess a **fractal character**, repeating its patterns at every level of existence.
3. **The Nadsoliton Concept:** The universe is conceptualized as a single, self-sustaining, non-dispersive wave packet—a **"Supersoliton" (Nadsoliton)**, where information tends towards the highest resonance, not the lowest energy.
4. **Resonant Structure:** Inspired by the Divine Name from the Book of Exodus 3:14: ***"I AM WHO I AM"***, the model incorporates **multi-octave resonant coupling** as the mechanism of self-organization, preventing decay into entropy.
5. **The 12-Octave Lattice:** Initial 3-octave models were expanded to a **12-octave structure**, inspired by the symbolic description of the Holy City's twelve foundation layers, which proved to be the mathematically necessary dimension for unifying all forces (Kissing Number in 3D).
6. **Access to Truth:** Since human consciousness is part of this informational substrate, the human mind has direct access to fundamental truths through wisdom and intuition, allowing for the "decoding" of reality.

Status discipline (no false pass)

This TeX document is the intended **Release-7 strict final documentation** target. It must not silently claim:

1. strict-core ToE closure,
2. global ToE closure,
3. kernel-alone/global QW-2191 discharge,
4. directed/sign-sensitive physical orientation datum in strict core,
5. host matching / Standard Model identification,
6. actual emergent observer closure,

unless such claims are explicitly exported at theorem level in the repo and are referenced here with their hard limits.

Reproducibility stamp (Release 7)

- **Repo root:** `/home/krzysiek/Pobrane/TOE/edison`
- **As-of date:** `2026-03-17`
- **Build smoke probe (PDF + strict dashboards):** `P707` (`fundamental_action_reconstruction/p707_cu`
- **Strict OS closure dashboard:** `P706` (`fundamental_action_reconstruction/p706_current_release_7_s`
- **Strict `T173` frontier dashboard:** `P708` (`fundamental_action_reconstruction/p708_current_strict_t1`
- **Previous-methodology survival audit:** `P711` (`fundamental_action_reconstruction/p711_current_st`

To reproduce the PDF and verify the current Release-7 strict dashboard statuses:

```
git rev-parse HEAD
python3 fundamental_action_reconstruction/
    p707_current_release_7_build_and_closure_smoke_probe.py
python3 fundamental_action_reconstruction/
    p711_current_strict_t173_previous_methodology_survival_and_global_gap_audit_probe
    .py
cat fundamental_action_reconstruction/generated/
    p707_current_release_7_build_and_closure_smoke_probe_summary.json
cat fundamental_action_reconstruction/generated/
    p711_current_strict_t173_previous_methodology_survival_and_global_gap_audit_probe_s
    .json
cat fundamental_action_reconstruction/generated/
    p706_current_release_7_strict_projective_operational_toe_os_closure_dashboard_probe
    .json
cat fundamental_action_reconstruction/generated/
    p708_current_strict_t173_frontier_dashboard_probe_summary.json
```

Contents

1 Ontology and scope

2 AX9 ontology discipline

The strict program is built under the ontology correction `AX9`:

- the nadsoliton itself is the primordial information of the universe in a solitonic state;
- there is no separate informational layer underneath the nadsoliton;
- the preferred internal order is:

$$\text{nadsoliton} \;\rightarrow\; \text{light} \;\rightarrow\; \text{matter} \;\rightarrow\; \text{emergent observer.}$$

In particular, we do not allow observer-facing quantities to silently back-propagate into the upstream definition of the nadsoliton core. Operationally, throughout this document, "light" is treated as the linearized eigenmode layer exported from the Ψ-sector Hessian instantiation (`F459`), and observer-facing readouts remain downstream (`P697`, `P699`).

3 Strict vs legacy kernel split discipline

The repo contains two historically distinct kernel objects (see `K1/K2/F2`):

$$K_{\text{legacy\_ont}}(d) := \alpha_{\text{geo}} \frac{\cos(\omega d + \phi)}{1 + \beta_{\text{tors}} d}, \qquad K_{\text{strict\_gate}}(d) := \frac{\cos(\omega d + \phi)}{1 + \beta d^{\eta}}.$$

By author decree (`S2`, 2026-03-09), the legacy kernel is retired to archival status; this document is strict-only. However, **no false pass** discipline remains: we do *not* silently transfer legacy physical-role identities (e.g. electroweak-angle or EM coupling slogans) onto the strict working kernel without an explicit bridge/role-transfer theorem.

On the current strict kernel-mode lane, the working strict tuple is the `QW-2049` gate-selected kernel with $(\omega, \phi, \beta, \eta) = (0.18575,\ 0.16250,\ 1.0,\ 1.8)$, used operationally to generate a frozen numeric carrier mixing matrix K_{total} on $\mathbb{Z}_{12}$ (`R14`).

4 What "closure" means in this document

We use the word "closure" in a deliberately scope-disciplined way.

Definition 4.1 (Operational ToE closure (strict projective OS)). *We say that the current repo state achieves* Operational ToE closure *in the strict* projective *scope if it exports a theorem-level operational support bundle sufficient to compute (from strict projective selector closure output data) the downstream artifacts recorded by the OS packet:*

$$\Lambda_{\text{OS}}^{(\text{proj})} := \big(\texttt{N680},\ \texttt{N692},\ \texttt{N693},\ \texttt{P694},\ \texttt{P696},\ \texttt{P697},\ \texttt{N699}\big),$$

with hard limits explicitly keeping kernel-alone/global *`QW-2191` discharge, directed sign-sensitive physical orientation, and ToE closure unclaimed.*

Remark 4.1 (OS v2 vs OS v3). *The minimal theorem-level OS support bundle used to discharge* Operational ToE closure *is the v2 packet (`N700/N701`). The v3 packet (`F705/P705/N705`) is a strict extension that additionally tracks the basis-invariant Hessian eigen-spectrum mass observable (`F704`). It does not strengthen any hard-limit claims.*

Remark 4.2 (Boundary discipline). *Operational ToE closure in the above sense is a closure of an* operational computability pipeline *(a ToE "OS"), not a claim of global physical identification (host matching) and not a claim of kernel-alone/global `QW-2191` discharge. It is compatible with leaving those stronger claims explicitly open.*

Theorem 4.1 (Operational ToE closure in strict projective OS scope (N701)). *On the current repo state, the repo exports a theorem-level strict projective operational OS support packet v2 (`N700`), hence discharges Operational ToE closure in strict projective OS scope (`N701`):*

$$\Lambda_{\text{OS}}^{(\text{proj})} = \big(\texttt{N680},\ \texttt{N692},\ \texttt{N693},\ \texttt{P694},\ \texttt{P696},\ \texttt{P697},\ \texttt{N699}\big),$$

with all hard limits explicit (no kernel-alone/global `QW-2191` discharge, no directed physical sign datum in strict core, no ToE closure, no actual emergent observer closure).

5 Strict core Lagrangian and emergence map

6 Single-kernel strict candidate (A11)

This chapter records the strict candidate "one-kernel" nadsoliton core Lagrangian as exported/packaged by `A11`. It is a definition/packaging object, not a claim that every coefficient family has already been strict-derived from the kernel alone.

6.1 Typed carrier and cyclic distance ($\mathbb{Z}_{12}$)

Let

$$I_{12} := \{0, 1, \ldots, 11\}, \qquad \mathbb{Z}_{12} := (I_{12}, + \bmod 12).$$

Define the cyclic octave distance on the ring:

$$d_{\text{cyc}}(i,j) := \min\Big((j-i) \bmod 12,\ (i-j) \bmod 12\Big) \in \{0, 1, \ldots, 6\}.$$

6.2 Strict working kernel and frozen mixing matrix

The strict working kernel is the later-pipeline gate-selected kernel (`QW-2049`):

$$K_{\text{strict\_gate}}(d) := \frac{\cos(\omega d + \phi)}{1 + \beta d^{\eta}}, \qquad (\omega, \phi, \beta, \eta) = (0.18575,\ 0.16250,\ 1.0,\ 1.8).$$

On the kernel-mode lane, the repo exports a frozen numeric symmetric mixing matrix $K_{\text{total}} \in \mathbb{R}^{12\times 12}$ on the same carrier (`QW-2118`), specialized into the canonical kernel channel by `R14`. In this document we only require the exported matrix entries (no additional kernel-to-matrix derivation is assumed here).

6.3 Fields and core Lagrangian density

Introduce a 12-component real field on spacetime:

$$\Psi(x) = (\psi_0(x), \ldots, \psi_{11}(x)) \in \mathbb{R}^{12},$$

and a real scalar order/coherence field:

$$\Phi(x) = \phi(x) \in \mathbb{R}.$$

Write the mixing potential using the exported K_{total}:

$$V_{\text{mix}}(\Psi) := \frac{1}{2} \sum_{\substack{i,j \in I_{12} \\ i \neq j}} (K_{\text{total}})_{ij}\, \psi_i\, \psi_j.$$

Write the local self-interaction potentials in the canonical template form (`QW-2163/2165/2166`):

$$V_{\Phi}(\phi) := \frac{1}{2} m_{\phi}^2\, \phi^2 + \frac{1}{4} \lambda_{\phi}\, \phi^4,$$

$$V_{\Psi}(\Psi) := \sum_{i=0}^{11} \left(\frac{1}{2} m_{\psi i}^2\, \psi_i^2\ +\ \frac{1}{4} g4_{\psi i}\, \psi_i^4\ +\ \frac{1}{6} g6_{\psi i}\, \psi_i^6 \right),$$

$$V_{\text{Y}}(\Psi, \phi) := \sum_{i=0}^{11} gY_i\, \phi^2\, \psi_i^2.$$

Then the nadsoliton core Lagrangian density is:

$$\mathcal{L}_{\text{core}} = \frac{1}{2}\, \partial_\mu \phi\, \partial^\mu \phi\ +\ \frac{1}{2} \sum_{i=0}^{11} \partial_\mu \psi_i\, \partial^\mu \psi_i\ -\ V_{\Phi}(\phi)\ -\ V_{\Psi}(\Psi)\ -\ V_{\text{Y}}(\Psi, \phi)\ -\ V_{\text{mix}}(\Psi).$$

6.4 Quadratic expansion, Ψ Hessian, and "mass" proxy meaning (strict, scope-limited)

Let (ϕ_*, Ψ_*) be a background configuration satisfying the stationarity condition for the potential part $V := V_\Phi + V_\Psi + V_\mathrm{Y} + V_\mathrm{mix}$:

$$\nabla_{\phi,\Psi} V(\phi_*, \Psi_*) = 0.$$

Write $(\phi, \Psi) = (\phi_*, \Psi_*) + (\delta\phi, \delta\Psi)$ and expand the potential to second order in $\delta\Psi$:

$$V(\phi_*, \Psi_* + \delta\Psi) = V(\phi_*, \Psi_*) \; + \; \frac{1}{2}\,\delta\Psi^\mathsf{T} H_\psi\,\delta\Psi \; + \; \cdots\,,$$

where $H_\psi \in \mathbb{R}^{12\times 12}$ is the Ψ-sector Hessian block (second derivatives) at the background point.

Hessian stencil from $\mathcal{L}_\mathrm{core}$ (canonical template `QW-2166`). Let $(v_\phi, v_{\psi 0}, \ldots, v_{\psi 11})$ denote a constant background vacuum. In the strict canonical action/Hessian export (`QW-2166`), the quadratic Ψ-sector Hessian block has the following stencil structure:

1. diagonal entries:
$$(H_\psi)_{ii} = m_{\psi i}^2 + 3\,g4_{\psi i}\,v_{\psi i}^2 + 5\,g6_{\psi i}\,v_{\psi i}^4 + 2\,gY_i\,v_\phi^2,$$
2. off-diagonal mixing entries ($i \neq j$):
$$(H_\psi)_{ij} = \frac{K_{ij} + K_{ji}}{2},$$
3. Ψ–Φ cross-block entries:
$$(H_{\psi\phi})_i = \frac{\partial^2 V}{\partial\psi_i\,\partial\phi}(v_\phi, v_\psi) = 4\,gY_i\,v_\phi\,v_{\psi i}.$$

This pins down what "light = linearized modes" means inside the program (eigenmodes of the Hessian around a vacuum), without asserting that the vacuum values or all coefficient families are already strict-derived from the kernel alone.

On the current diagonal/local strict-derived lane, the repo exports one explicit numeric value instantiation (`F459`) of the form:

$$H_\psi := K_\mathrm{total} + \big(m_0^2 I + D_\mathrm{local\_residual}\big),$$

with $m_0^2 I$ the exported host scalar floor embedding and $D_\mathrm{local\_residual}$ an exported diagonal/local strict-derived residual profile.

In strict scope we treat the quadratic coefficient $\delta\Psi^\mathsf{T} H_\psi \delta\Psi$ as defining a *dimensionless* "m^2 proxy" layer: for any nonzero vector u, the Rayleigh quotient $u^\mathsf{T} H_\psi^{(s)} u/(u^\mathsf{T} u)$ is a well-defined quadratic proxy, where $H_\psi^{(s)} = \frac{1}{2}(H_\psi + H_\psi^\mathsf{T})$. This does *not* yet constitute a physical unit identification, and it does not bypass the strict-core selector obstruction `QW-2191`; it is an operational computability meaning only (packaged as a scope-limited meaning theorem `N703`).

Theorem 6.1 (Quadratic mass-proxy meaning definition (N703)). *On the current repo state, the proxy quantities exported by `P694` and `P696` are interpreted strictly as* dimensionless quadratic coefficients *of the exported diagonal/local Ψ-sector operator instantiation (`F459`): they are Rayleigh/Ritz-type quadratic forms of $H_\psi^{(s)}$ evaluated on exported projective selector rays (from `F672`) and deterministic in-plane complements. No physical-unit map, no Standard Model host identification, and no diagonalization claim is discharged.*

Hard limits: *no directed physical sign datum, no kernel-alone/global `QW-2191` discharge, and no ToE closure are claimed.*

7 Light → matter → observer ladder

`A11` fixes a disciplined emergence ladder. On the current repo state, the strongest honest statements along that ladder are:

1. **Light (discharged as computable instantiation):** the repo exports a strict-derived diagonal/local value instantiation of the Ψ-sector Hessian H_ψ and its full real eigensystem (`F459`), plus sign-gauge-invariant eigenprojectors (`F460`). Mixing across Fourier pair planes is explicitly audited and packaged as a value-instantiation boundary (`N505`); therefore no "pair-plane diagonalization" shortcut is allowed.

2. **Matter (programmatic, not closed):** matter is intended to arise from nonlinear structure in the same core Lagrangian (local quartic/sextic families and Yukawa-type couplings), but the repo does not yet export a strict derivation that these families are kernel-alone consequences or a closed host identification map.

3. **Emergent observer (operational OS closure, not actual observer closure):** from the strict projective selector closure output on C_{v1} (`N680`, exported as `F672`), the repo exports a projective observer-limit readout (`P697`) and a fully computable downstream projective chain under projector pushforward semantics $P \mapsto MPM^{\mathsf{T}}$ (`P699`/`N699`). This suffices for *Operational ToE closure* in strict projective OS scope (`N701`), while *actual emergent observer closure* remains explicitly open.

8 Selector closure on C_{v1}

9 Strict configuration space C_{v1} and the $\pi_1(C_{v1}) \cong \mathbb{Z}_2$ witness (F306)

The strict selector program is typed over an explicit strict configuration-space object `C_v1_void_configuration_sp` (exported by `F306`). In strict scope, we only use the following minimal mathematical content of that export:

Definition 9.1 (Model declaration for C_{v1} (no physics claims))**.** *On the declared local protected sector, interpret the configuration space as the based mapping space model:*

$$C_{v1} := \mathrm{Map}_*^{\deg=1}\big(S^3, \mathrm{SU}(2)\big)\,.$$

This is the standard finite-energy compactification discipline: $\mathbb{R}^3 \cup \{\infty\} \cong S^3$ *with boundary condition* $U(\infty) = 1 \in \mathrm{SU}(2)$*, so a configuration is a based map* $S^3 \to \mathrm{SU}(2)$*.*

Proposition 9.1 (Topological witness $\pi_1(C_{v1}) \cong \mathbb{Z}_2$ (F306))**.** *On the above model, the fundamental group satisfies:*

$$\pi_1(C_{v1}) \;\cong\; \mathbb{Z}_2,$$

and `F306` exports an explicit generator label $\gamma^{(v1)}_{\pi_1}$ *for the nontrivial loop class.*

Proof sketch (classical algebraic topology). Using the loop–suspension adjunction,

$$\Omega\,\mathrm{Map}_*(S^3, S^3) \;\simeq\; \mathrm{Map}_*(\Sigma S^3, S^3) \;=\; \mathrm{Map}_*(S^4, S^3),$$

hence

$$\pi_1\big(\mathrm{Map}_*(S^3, S^3)\big) = \pi_0\big(\Omega\,\mathrm{Map}_*(S^3, S^3)\big) \cong \pi_0\big(\mathrm{Map}_*(S^4, S^3)\big) = [S^4, S^3] = \pi_4(S^3) \cong \mathbb{Z}_2,$$

where the final identification $\pi_4(S^3) \cong \mathbb{Z}_2$ is a classical homotopy-group fact. This witness is strictly below any physical FR-sign quantization claims and is used only as a typed topology availability result. □

10 Real Fourier carrier on $\mathbb{Z}_{12}$ and the $O(2)$ obstruction (QW-2191/N465)

The strict selector program is built on a typed 12-slot carrier (`F329`), and its canonical degeneracy/uniqueness obstruction `QW-2191` is most transparently expressed in the real Fourier basis on $\mathbb{Z}_{12}$.

10.1 Explicit orthonormal real Fourier basis

Index the carrier by $x \in I_{12} = \{0, \dots, 11\}$ and identify $\mathbb{R}^{12}$ with real-valued functions on $\mathbb{Z}_{12}$. Define:

$$e_0(x) := \frac{1}{\sqrt{12}}, \qquad e_6(x) := \frac{(-1)^x}{\sqrt{12}}.$$

For each $m = 1, \dots, 5$, define the real Fourier pair-plane vectors:

$$c_m(x) := \sqrt{\frac{2}{12}}\cos\left(\frac{2\pi m x}{12}\right), \qquad s_m(x) := \sqrt{\frac{2}{12}}\sin\left(\frac{2\pi m x}{12}\right).$$

Then $\{e_0, e_6, c_1, s_1, \dots, c_5, s_5\}$ is an orthonormal basis of $\mathbb{R}^{12}$ and induces the canonical orthogonal decomposition:

$$\mathbb{R}^{12} \;=\; \mathrm{span}\{e_0\} \;\oplus\; \mathrm{span}\{e_6\} \;\oplus\; \bigoplus_{m=1}^{5} \mathrm{span}\{c_m, s_m\}.$$

We denote the m-th pair plane by:

$$V_m := \mathrm{span}\{c_m, s_m\}.$$

10.2 Why cyclic-distance kernels cannot select a unique axis inside V_1

Let $T : \mathbb{R}^{12} \to \mathbb{R}^{12}$ be the cyclic shift operator:

$$(Tf)(x) := f(x - 1 \bmod 12).$$

If the host kernel K_{total} is constructed as a cyclic-distance profile (`QW-2118`), then K_{total} commutes with T. On each pair plane V_m, the restriction $T|_{V_m}$ is a genuine planar rotation by angle $2\pi m/12$ on the basis (c_m, s_m). Therefore, any real symmetric operator commuting with T restricts to a scalar multiple of the identity on V_m.

Theorem 10.1 (Host isotropy on pair1 (N465))**.** *For the strict cyclic-distance host operator $A := K_{\text{total}} + m_0^2 I$ (`QW-2118`+`QW-2124`), the restriction to the pair1 plane $V_1 = \mathrm{span}\{c_1, s_1\}$ is isotropic:*

$$A|_{V_1} = \lambda_1 I_2.$$

Therefore, the host operator cannot supply an $O(2)$-cut inside pair1 and cannot canonically distinguish c_1 from s_1 (nor any rotated basis).

Proof sketch (from cyclic equivariance). Because K_{total} depends only on cyclic distance, it commutes with the cyclic shift T. The scalar floor $m_0^2 I$ also commutes with T, hence A commutes with T. On $V_1 = \mathrm{span}\{c_1, s_1\}$, the action of T is a planar rotation by angle $2\pi/12 = \pi/6$ in the basis (c_1, s_1). Any linear map commuting with a nontrivial planar rotation has the form $aI_2 + bJ$ with $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Since A is symmetric, its restriction $A|_{V_1}$ is symmetric, forcing $b = 0$, hence $A|_{V_1} = aI_2$ for some scalar $a = \lambda_1$. □

Remark 10.1 (Meaning: the `QW-2191` uniqueness obstruction is structural)**.** *The above isotropy is a concrete instance of the general strict obstruction `QW-2191`: kernel-alone translation-invariant data leave an $O(2)$ family of basis choices inside each degenerate pair plane V_m. This is why Release 7 closure statements are deliberately phrased in* projective (ray-level) *scope unless an explicit symmetry-breaking/selector source is tracked.*

11 Global atlas and transition objects (T170)

In the strict selector program, the configuration space domain is the declared strict object C_{v1}. On the current repo state, the global chart/transition infrastructure required for a global selector discussion is exported as objects:

- a global selector atlas object on C_{v1} (`F469`, packaged by `N515`),
- a global selector transition (gluing) object on C_{v1} (`F469`, packaged by `N515`).

In the current strict global-typing export (`F469`), the chart domains are declared as a global cover:

$$U_{\mathrm{pair}_1} = \cdots = U_{\mathrm{pair}_5} = C_{v1}, \qquad U_i \cap U_j = C_{v1}.$$

This declaration upgrades the earlier lane-scoped "exported-artifact overlap" wording into an explicit C_{v1}-typed atlas/transition object class discharge (T170), but it does *not* add any physical claim and does *not* promote operator-level cocycle/groupoid identities (`N512` boundary).

The atlas uses five chart labels pair_m ($m = 1, \ldots, 5$), corresponding to the five real Fourier pair planes $V_m = \mathrm{span}\{c_m, s_m\} \subset \mathbb{R}^{12}$. Each chart exports a local rank-1 projector operator A_m (a ray/state projector) on that plane. At projector level, if u_m is a representative vector in the plane, then

$$A_m = \frac{u_m u_m^\top}{u_m^\top u_m}, \qquad u_m \sim -u_m \quad \Rightarrow \quad A_m \text{ is sign-gauge-invariant.}$$

The transition object exports explicit orthogonal chart-transport operators O_{ij} between the chart planes (axis-only/projector-level transport). Projector transport is by conjugation:

$$A_j \;=\; O_{ij}\, A_i\, O_{ij}^\top.$$

Cocycle discipline: the repo keeps a strict boundary that cocycle/path-independence is a *projector-section-level* statement only; operator-level identities $O_{jk}O_{ij} = O_{ik}$ are explicitly *not* promoted in strict core (`N512`).

11.1 Exported JSON-level interfaces (F469)

This section makes the **exact exported object interfaces** explicit, so that the strict global selector infrastructure can be audited without opening the repo.

Definition 11.1 (Exported atlas interface: `SelectorAtlas_global_C_v1_strict_v1` (F469))**.** *The atlas object is exported as the JSON artifact* `fundamental_action_reconstruction/generated/selector_atlas_g` *Its minimal strict interface consists of:*

1. `domain`*: a reference to the declared strict configuration space object (*`.../c_v1_void_configuration_space_i` *and the explicit global-cover declaration* $U_{\mathrm{pair}_1} = \cdots = U_{\mathrm{pair}_5} = C_{v1}$*;*
2. `charts`*: a map over chart labels* pair_m *containing:*
   - `carrier_plane` *(the Fourier pair plane* $V_m = \mathrm{span}\{c_m, s_m\}$*),*
   - `local_operator.ref` *(a JSON ref for the local rank-1 projector operator* A_m*);*
3. `chart_domains`*: a typed chart-domain declaration on* C_{v1} *for each chart label;*
4. `overlap_domains`*: an explicit overlap-domain declaration for each unordered pair* $\mathrm{pair}_i \cap \mathrm{pair}_j$ *(in this export:* C_{v1} *by global-domain choice);*
5. `transitions`*: a complete-graph edge map* $\mathrm{pair}_i \to \mathrm{pair}_j$ *containing:*

- `operator_ref`: *a JSON ref for the orthogonal transport operator* O_{ij},
- *one parameter field* `alpha..` *recording the exported angle value (mod* π *or* 2π*).*

Hard limits: *these are declared-typing atlas objects; they do not promote an operator-level groupoid or strict-core physical sign convention.*

Definition 11.2 (Exported transition interface: `SelectorTransition_global_C_v1_strict_v1` (F469))**.** *The transition object is exported as* `fundamental_action_reconstruction/generated/selector_transition_glo` *Its minimal strict interface consists of:*

1. `transition_operators`: *the same complete-graph edge map of operator refs* O_{ij} *and angle fields;*
2. `gluing_laws_ref`: *a reference to the exported five-chart projector-section cocycle audit object* `.../a_12345_pair12345_chart_glued_orientation_projector_operator_section_strict_core_v2.jso`

The `gluing_laws_ref` *object records explicit* projector-section-level *transport laws of the form* $A_j = O_{ij} A_i O_{ij}^{\top}$ *and triple-overlap cocycle/path-independence audits (max-entry residuals).* Hard limits: *this remains* projector-section-level *only and does not promote to operator-level identities (*`N512`*).*

11.2 Exported transition angles (F469)

For convenience, Table 1 records the exported transition angles (in radians) from `selector_transition_global_c_` Numerically, all exported values are consistent (within floating-point tolerance) with multiples of $\pi/2$.

Table 1: Exported transition angles on the five-chart selector atlas/transition (F469).

Edge	Exported field	Value (rad)
$\text{pair}_1 \to \text{pair}_2$	`alpha12_mod_2pi`	1.570796326794897
$\text{pair}_1 \to \text{pair}_3$	`alpha13_mod_pi`	1.570796326794897
$\text{pair}_1 \to \text{pair}_4$	`alpha14_mod_pi`	1.570796326794896
$\text{pair}_1 \to \text{pair}_5$	`alpha15_mod_pi`	0.0
$\text{pair}_2 \to \text{pair}_3$	`alpha23_mod_pi`	0.0
$\text{pair}_2 \to \text{pair}_4$	`alpha24_mod_pi`	3.141592653589793
$\text{pair}_2 \to \text{pair}_5$	`alpha25_mod_pi`	1.570796326794897
$\text{pair}_3 \to \text{pair}_4$	`alpha34_mod_pi`	0.0
$\text{pair}_3 \to \text{pair}_5$	`alpha35_mod_pi`	1.570796326794898
$\text{pair}_4 \to \text{pair}_5$	`alpha45_mod_pi`	1.570796326794898

11.3 Projective state, output operator, and closure interfaces (F470/F660/F672)

The strict projective closure relies on three exported global objects whose interfaces are intentionally simple.

Definition 11.3 (Projective state interface: `SelectorState_global_C_v1_projective_strict_v1` (F470))**.** *The projective selector state is exported as* `fundamental_action_reconstruction/generated/selector_` *Its strict interface consists of:*

1. `global_atlas_ref` *and* `global_transition_ref` *pointing to the* `F469` *objects;*
2. `charts`: *the per-chart local projector-operator refs* A_m*;*
3. `projector_section_ref`: *a reference to the five-chart glued projector-section cocycle audit object* `.../a_12345_pair12345_chart_glued_orientation_projector_operator_section_strict_core_v2.jso`

4. *`state_type`: a declaration that the state is a* projective ray state *(sign gauge $u \sim -u$ at state level).*

Definition 11.4 (Output operator interface: `SelectorOutputOperator_global_C_v1_seed_v1_promoted_strict` (F660)). *The global output operator is exported as* `fundamental_action_reconstruction/generated/selector_ou` *It contains the promoted seed-v1 output map $O_{\mathrm{sel}} : Q_{\mathrm{sel}} \to Q_{\mathrm{out}}$ in a fixed declared basis.*

Definition 11.5 (Projective closure interface: `SelectorClosure_global_C_v1_projective_strict_v1` (F672)). *The projective closure object is exported as* `fundamental_action_reconstruction/generated/selector_` *Its strict interface consists of:*

1. *references to the atlas, transition, projective state, and output operator objects;*
2. *`output_observable`: the resulting global chart-independent rank-1 output projector matrix B_{out} on $Q_{\mathrm{out}} = \mathrm{span}\{o_+, o_-\}$;*
3. *`well_definedness_certificate`: a numeric chart-independence certificate with fields: `criterion`, `tolerance`, `max_abs_diff_to_reference_chart_output_projector`, `certificate_pass`.*

Hard limits: *projective closure is the invariant content; no directed physical sign convention is promoted.*

12 Seed-v1 strict-core internal selector-source chain (S_sel_int → E_orient → R_sel → O_sel)

The projective closure object `F672` is constructed using a strict-core internal selector-source seed chain (seed-v1) that provides, in strict scope:

- a strict-core source object for the `S_sel_int` step,
- an admissible orientation datum on `pair1` (`E_orient`),
- a strict reduction operator into the selector-sector basis `Q_sel_v1` (`R_sel`),
- a strict output operator into the output-sector basis `Q_out_v1` (`O_sel`).

All these objects are strictly below any kernel-alone/global `QW-2191` discharge and below any directed/sign-sensitive physical orientation datum.

12.1 Minimal admissibility contract for the source step (F34)

The strict program treats `S_sel_int` as an internal *source* step. The minimal admissibility contract for a strict-core source object to count as `S_sel_int` (in the sense of `F34`) is:

1. genuinely new strict-core source object (not a reuse of control/axiom artifacts),
2. carrier-typed enough for later export (so that `E_orient` can be defined from it),
3. source-seed only (not a hidden discharge of downstream steps),
4. strict-core only,
5. non-substitutive (no silent legacy-to-strict kernel identification),
6. selector-acceptance independent (non-strict acceptance does not count as source construction),
7. future-bridge compatible (stable enough for later strict continuation).

Theorem 12.1 (Admissible strict-core source object for `S_sel_int` exists (N676)). *On the current repo state, the strict core exports one admissible source object satisfying the above `F34` contract for the `S_sel_int` source step:*

`S_sel_int_strict_core_source_object_v1`.

Hard limits: *this is not yet a strict-core selector closure claim and does not discharge kernel-alone/global `QW-2191` or ToE closure.*

12.2 Explicit strict-core payload weights exported by the source object (F647)

The constructed strict-core source object exported by the seed-v1 realization attempt (`F647`) includes a minimal, explicitly enumerated carrier-level payload on $\mathbb{Z}_{12}$:

- `ord_z12_by_x`: the group element order $\mathrm{ord}_{\mathbb{Z}_{12}}(x)$,
- `w_ref_unnormalized_by_x`: a strict-derived element-order reference weight profile,
- `w_break_by_x`: a reflection-breaking weight profile (used only as a convention-layer sign anchor; no physical sign datum).

Element order on $\mathbb{Z}_{12}$. For $x \in \mathbb{Z}_{12}$ (additive notation), define:

$$\mathrm{ord}_{\mathbb{Z}_{12}}(x) := \min\{n \geq 1 : \ nx \equiv 0 \pmod{12}\} \ = \ \frac{12}{\gcd(12, x)}.$$

This is $\mathrm{Aut}(\mathbb{Z}_{12})$-invariant.

Strict-derived reference weight. Using the strict-derived Shannon asymmetry constant $\alpha_{\mathrm{geo}} := 4 \ln 2$ (`F309`/`N420`), define the unnormalized order-reference weight:

$$w_{\mathrm{ref}}^{\mathrm{unnorm}}(x) := \exp\big(-\alpha_{\mathrm{geo}} \ \mathrm{ord}_{\mathbb{Z}_{12}}(x)\big) = 2^{-4\,\mathrm{ord}_{\mathbb{Z}_{12}}(x)}.$$

Reflection-breaking weight (strict convention scope). Define:

$$w_{\mathrm{break}}(x) := -\, w_{\mathrm{ref}}^{\mathrm{unnorm}}(x) \ \sin\!\left(\frac{2\pi x}{12}\right).$$

This weight is *odd* under reflection $x \mapsto -x$ and therefore breaks the reflection symmetry on the carrier. It is not $\mathrm{Aut}(\mathbb{Z}_{12})$-invariant and is treated as a convention-layer sign anchor only.

Lemma 12.2 (Direction-free weights cannot distinguish a sine-axis sign (N517/N518)). *Let $w : \mathbb{Z}_{12} \to \mathbb{R}$ be* reflection-even*:*

$$w(x) = w(-x) \quad \text{for all } x \in \mathbb{Z}_{12}.$$

Then for every Fourier sine basis vector s_m $(m = 1, \ldots, 5)$ one has

$$\langle w, s_m \rangle := \sum_{x \in \mathbb{Z}_{12}} w(x)\, s_m(x) = 0.$$

In particular, a deterministic sign rule of the form

$$u \ \mapsto \ \mathrm{sign}\big(\langle w, u \rangle\big)\, u$$

cannot distinguish u from $-u$ when the exported axis representative lies exactly on a sine axis (e.g. $u = \pm s_1$ on pair1*).*

Moreover, since $-1 \in \mathrm{Aut}(\mathbb{Z}_{12})$, any $\mathrm{Aut}(\mathbb{Z}_{12})$-invariant weight profile is necessarily reflection-even. Therefore no direction-free (Aut-invariant) reference-weight family can supply a strict-core sign-distinction scalar on a pure sine axis.

Proof. Because $s_m(-x) = -s_m(x)$, we may pair terms x and $-x$:

$$w(x)s_m(x) + w(-x)s_m(-x) = w(x)s_m(x) - w(-x)s_m(x) = (w(x) - w(-x))s_m(x) = 0.$$

The fixed points $x = 0$ and $x = 6$ satisfy $s_m(0) = s_m(6) = 0$, so they contribute nothing. Hence the total sum vanishes. For the Aut-invariance statement, apply the automorphism $x \mapsto -x$ and invariance of w under $\mathrm{Aut}(\mathbb{Z}_{12})$. □

Exported numeric instance (F647 payload). The exported payload values on the current repo state are:

x	$\mathrm{ord}_{\mathbb{Z}_{12}}(x)$	$w_{\mathrm{ref}}^{\mathrm{unnorm}}(x)$	$w_{\mathrm{break}}(x)$
0	1	$6.2500000000000000 \times 10^{-2}$	0
1	12	$3.5527136788005110 \times 10^{-15}$	$-1.7763568394002552 \times 10^{-15}$
2	6	$5.9604644775390720 \times 10^{-8}$	$-5.1619136559035774 \times 10^{-8}$
3	4	$1.5258789062500007 \times 10^{-5}$	$-1.5258789062500007 \times 10^{-5}$
4	3	$2.4414062500000016 \times 10^{-4}$	$-2.1143198334581037 \times 10^{-4}$
5	12	$3.5527136788005110 \times 10^{-15}$	$-1.7763568394002568 \times 10^{-15}$
6	2	$3.9062500000000010 \times 10^{-3}$	$-4.7837765591693490 \times 10^{-19}$
7	12	$3.5527136788005110 \times 10^{-15}$	$1.7763568394002544 \times 10^{-15}$
8	3	$2.4414062500000016 \times 10^{-4}$	$2.1143198334581029 \times 10^{-4}$
9	4	$1.5258789062500007 \times 10^{-5}$	$1.5258789062500007 \times 10^{-5}$
10	6	$5.9604644775390720 \times 10^{-8}$	$5.1619136559035800 \times 10^{-8}$
11	12	$3.5527136788005110 \times 10^{-15}$	$1.7763568394002570 \times 10^{-15}$

The same payload export records the strict scalar sign-anchor witness:

$$\langle w_{\mathrm{break}}, s_1 \rangle \approx -3.9681792259571596 \times 10^{-4},$$

where s_1 is the real Fourier sine basis vector of the pair1 plane (Section 10). This is used downstream only to define deterministic convention-layer sign fixing rules (e.g. `T175`/`F690`) and is not promoted into strict-core physics.

12.3 Orientation datum on pair_1 (N546)

Theorem 12.3 (Admissible orientation export (N546)). *Within the current repo state, the exported strict-core source object*

`S_sel_int_strict_core_source_object_v1`

exports one admissible orientation datum on the ***`pair1`*** *frame* $\mathrm{span}\{c_1, s_1\}$*:*

`E_orient_s_sel_int_source_object_v1`.

Hard limits: *this does not promote a directed physical sign datum and does not by itself discharge strict-core selector closure.*

12.4 Selector-sector and output-sector operators (N549)

Let the selector-sector and output-sector carrier spaces be fixed as two-dimensional real spaces with declared bases:

$$Q_{\mathrm{sel}} := \mathrm{span}\{q_+, q_-\}, \qquad Q_{\mathrm{out}} := \mathrm{span}\{o_+, o_-\}.$$

Theorem 12.4 (Seed-v1 selector output operator exists (N549)). *Within the current repo state, the seed-v1 lane exports an admissible strict-core selector output operator*

$$O_{\mathrm{sel}} : \; Q_{\mathrm{sel}} \to Q_{\mathrm{out}},$$

packaged as the exported object `O_sel_s_sel_int_source_object_v1`, derived from an admissible reduction operator `R_sel_s_sel_int_source_object_v1`. Hard limits: *no strict-core selector closure, no kernel-alone/global `QW-2191` discharge, and no ToE closure are claimed.*

12.5 Global promotion to C_{v1}-typed operator families (N551/N552)

Theorem 12.5 (Global reduction operator family promotion (N551)). *On the current repo state, the repo exports a global (C_{v1}-typed) selector reduction operator family $R_{\mathrm{sel}}(\mathrm{pair}_m) : \mathrm{pair}_m \to Q_{\mathrm{sel}}$ promoted from the seed-v1 chain on `pair1` to all charts $\{\mathrm{pair}_1, \ldots, \mathrm{pair}_5\}$, with overlap transport consistency kept at projector/section level (up to residual sign gauge).* Hard limits: *no strict-core selector closure and no global `QW-2191` discharge are claimed.*

Theorem 12.6 (Global output operator and chartwise channels (N552)). *On the current repo state, the repo exports a global (C_{v1}-typed) selector output operator $O_{\mathrm{sel}} : Q_{\mathrm{sel}} \to Q_{\mathrm{out}}$ and the induced chartwise output channels*

$$Y_{\mathrm{sel}}(\mathrm{pair}_m) := O_{\mathrm{sel}} \circ R_{\mathrm{sel}}(\mathrm{pair}_m) \qquad (m = 1, \ldots, 5),$$

with overlap transport consistency kept explicit up to residual sign gauge. Hard limits: *no strict-core selector closure, no kernel-alone/global `QW-2191` discharge, and no ToE closure are claimed.*

13 Projective closure (T172/T173 projective)

The global projective selector closure is exported as the object

`SelectorClosure_global_C_v1_projective_strict_v1` (`F672`),

with theorem-level discharge `N680`.

In projective scope, the selector state is a ray (projector/span semantics), and the closure observable is an output-space rank-1 projector computed chartwise and then certified as chart-independent:

$$B_{\mathrm{out}}(\mathrm{pair}_m) \;:=\; Y_{\mathrm{sel}}(\mathrm{pair}_m)\, A_m\, Y_{\mathrm{sel}}(\mathrm{pair}_m)^{\mathsf{T}} \in \mathbb{R}^{2\times 2},$$

in the fixed output basis (o_+, o_-). The chartwise output channel $Y_{\mathrm{sel}}(\mathrm{pair}_m) : \mathrm{pair}_m \to Q_{\mathrm{out}}$ is exported as a promoted seed-v1 chain object (`F660`) by:

$$Y_{\mathrm{sel}}(\mathrm{pair}_m) \;:=\; O_{\mathrm{sel}} \circ R_{\mathrm{sel}}(\mathrm{pair}_m),$$

where $R_{\mathrm{sel}}(\mathrm{pair}_m)$ is the exported chartwise selector reduction operator family and $O_{\mathrm{sel}} : Q_{\mathrm{sel}} \to Q_{\mathrm{out}}$ is the exported seed-v1 selector output operator (both promoted to global C_{v1}-typed objects in the strict program). Residual $\mathbb{Z}_2$ sign handling is tracked at the convention layer and is not promoted to a physical sign datum.

`F672` exports that all chartwise $B_{\mathrm{out}}(\mathrm{pair}_m)$ agree (within tolerance) and records the resulting global output projector:

$$B_{\mathrm{out}} = \begin{pmatrix} 1 & -9.149258638450154 \times 10^{-16} \\ -9.149258638450154 \times 10^{-16} & 8.370893363325476 \times 10^{-31} \end{pmatrix} \approx \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

so numerically the global output projector is the pure o_+ ray (machine-level residuals only).

Remark 13.1 (Explicit well-definedness certificate (F672)). *The strict projective closure export `F672` includes an explicit* well-definedness certificate *of chart-independence at projector level. Concretely, for a fixed reference chart (taken as `pair1` in the export), define the max-entry difference*

$$\Delta_m := \max_{a,b\in\{+,-\}} \left| \left(B_{\text{out}}(\text{pair}_m) - B_{\text{out}}(\text{pair}_1) \right)_{ab} \right| .$$

`F672` certifies:

$$\max_{m=1,\dots,5} \Delta_m \le 10^{-12}, \qquad \max_m \Delta_m \approx 7.27 \times 10^{-16},$$

and marks the certificate as passed. This is sufficient for projective/ray-level *closure scope and does not upgrade to any operator-level groupoid identity (`N512` boundary).*

This is a *projective* closure statement only: it is insensitive to sign lifts $u_m \mapsto -u_m$ and does not claim any directed/sign-sensitive physical orientation datum.

Theorem 13.1 (Projective strict-core selector closure (N680)). *On the current repo state, there exists one explicit global projective selector closure outcome on C_{v1} (`F672`) derived from exported strict-core objects, whose closure observable on $Q_{\text{out}} = \text{span}\{o_+, o_-\}$ is a well-defined chartwise-glued rank-1 output projector. Therefore, strict-core selector closure is discharged in* projective (ray-level) scope *on C_{v1}.*

Hard limits: *this does not discharge kernel-alone/global `QW-2191`, does not promote a directed/sign-sensitive physical orientation datum, and does not claim ToE closure.*

14 Directed lifts as gauge/convention layer

Beyond the projective closure object, the repo also exports directed/vector-level representatives and oriented sign-lift data as an explicit *convention layer* (gauge) on top of the projective/ray-level closure. The strict no-false-pass boundary is:

- directed lifts are tracked as gauge/convention data and are not a free strict-core physical sign datum;
- the projective output ray is the invariant content.

Concretely, `N692` packages that the directed closure output (with its tracked output sign-lift) induces the same rank-1 output projector as the projective closure output projector of `F672`. `N693` packages the gauge covariance of the directed output sign-lift under chartwise sign relifts.

Theorem 14.1 (Projective/directed closure output-ray invariance (N692)). *Let B_{out} denote the exported projective closure output projector on $Q_{\text{out}} = \text{span}\{o_+, o_-\}$ (`F672`). For each exported directed closure representative (in its declared scope), let $q_{\text{out}}^{(\text{dir})} \in \mathbb{R}^2$ be the directed output vector (with explicit sign-lift convention). Then the induced rank-1 projector*

$$\Pi\left(q_{\text{out}}^{(\text{dir})}\right) := \frac{q_{\text{out}}^{(\text{dir})} q_{\text{out}}^{(\text{dir})\mathsf{T}}}{q_{\text{out}}^{(\text{dir})\mathsf{T}} q_{\text{out}}^{(\text{dir})}}$$

agrees with B_{out} (within the exported numeric tolerance audits). Therefore, the strict global closure outcome on Q_{out} is projectively invariant across the exported directed lifts.

Hard limits: *this does not promote a directed/sign-sensitive physical orientation datum in strict core and does not discharge kernel-alone/global `QW-2191`.*

Theorem 14.2 (Chartwise $\mathbb{Z}_2$ gauge covariance of output sign-lift (N693))**.** *Let* $s^{\text{prem}}_{\text{out}}(\text{pair}_m) \in \{\pm 1\}$ *denote the exported output sign-lift for the premise-based directed closure on chart* pair_m*, and let* $t(\text{pair}_m) \in \{\pm 1\}$ *be the exported chartwise sign relift mapping the premise-based directed representative to the exported sign-fixed directed representative. Then the sign-fixed directed closure output sign-lift satisfies, for* $m = 1, \ldots, 5$*,*

$$s^{\text{fix}}_{\text{out}}(\text{pair}_m) = t(\text{pair}_m)\, s^{\text{prem}}_{\text{out}}(\text{pair}_m).$$

Therefore, the directed closure output sign-lift is a tracked $\mathbb{Z}_2$ *gauge datum and does not upgrade to a strict physical sign.*

Hard limits: *no directed physical orientation datum is claimed; no kernel-alone/global* `QW-2191` *discharge is claimed.*

On the current repo state, the directed layer is further stabilized by explicit convention-scoped sign data:

- oriented overlap-edge sign-lifts (`T174`: `F688`/`P688`/`N688`), with gauge-equivalence packaged (`N689`),
- deterministic chart sign fixing (0-cochain) from strict-core payload weights (`T175`: `F690`/`P690`/`N690`),
- a corresponding sign-fixed directed closure export (`F692`) that still induces the same projective output ray (by `N692`).

All of these remain *convention-layer* artifacts and do not upgrade any directed sign into strict-core physics.

Explicit $\mathbb{Z}_2$ cochain language for sign lifts (T174/T175)

Let the chart set be $\mathcal{C} = \{\text{pair}_1, \ldots, \text{pair}_5\}$ and let $\mathcal{E}$ be the set of 10 overlap edges (the complete graph on $\mathcal{C}$). For each edge $(i, j) \in \mathcal{E}$, the axis-only transition representative exports an operator $O^{(\text{axis})}_{ij}$ (rotation α_{ij} mod π).

Definition 14.1 (Edge sign lift (convention layer))**.** *An oriented edge sign lift is a* $\mathbb{Z}_2$*-valued* 1*-cochain* $s : \mathcal{E} \to \{\pm 1\}$ *defining oriented transitions:*

$$O^{(\text{oriented})}_{ij} := s_{ij}\, O^{(\text{axis})}_{ij}.$$

Definition 14.2 (Chart sign relift (gauge))**.** *A chartwise sign relift is a* $\mathbb{Z}_2$*-valued* 0*-cochain* $t : \mathcal{C} \to \{\pm 1\}$ *acting on directed representatives by:*

$$u_i \mapsto t_i u_i.$$

It induces the gauge action on edge signs:

$$s_{ij} \mapsto s'_{ij} := t_j\, s_{ij}\, t_i.$$

In this language, `T174` exports explicit oriented edge sign-lifts s_{ij} that remove sign flips for declared directed representatives (full-edge coherence), and `T175` exports one explicit deterministic chart sign-fix t_i derived from strict-core payload weights (Section 12.2). Both remain below any physical sign datum claim.

Exported instance (Release 7). On the current repo state, the deterministic chart sign-fixing 0-cochain exported by `F690` is:

$$(t_{\text{pair1}}, t_{\text{pair2}}, t_{\text{pair3}}, t_{\text{pair4}}, t_{\text{pair5}}) = (-1, -1, -1, +1, -1).$$

Anchored to that sign-fixed directed representative, the oriented edge sign-lift exported by `F691` flips exactly two overlap edges:

$$s_{\text{pair2}\to\text{pair4}} = -1, \qquad s_{\text{pair1}\to\text{pair5}} = -1,$$

with all remaining overlap edges having $s_{ij} = +1$ (convention scope; no physical sign claim).

15 Open boundary: kernel-alone/global QW-2191 discharge

Theorem 15.1 (Post-`T172` strict frontier boundary (N679)). *After the repo exports global selector closure observables on C_{v1} in both:*

- *projective scope (`F672`, packaged by `N672/N673/N674`), and*
- *directed premise-tracked scope (`F677`, packaged by `N677/N678`), with the raw output obstruction boundary recorded by `N675`,*

the repo packages the remaining strict frontier as a theorem-level boundary statement (`N679`):

1. *projective strict-core selector closure is discharged (`N680`),*
2. *but kernel-alone/global `QW-2191` discharge remains explicitly unclaimed,*
3. *and no directed/sign-sensitive physical orientation datum is promoted into strict core beyond premise-tracked sign-lift conventions.*

Hard limits: *`N679` does not claim kernel-alone/global `QW-2191` discharge, any directed physical sign datum in strict core, or ToE closure.*

`QW-2191` remains a real strict-core uniqueness obstruction: kernel-alone translation-invariant data leaves an $O(2)$ basis freedom inside each degenerate Fourier pair plane.

The exported global closure in this document is therefore deliberately in *projective* scope (ray/projector semantics), and the repo keeps kernel-alone/global `QW-2191` discharge explicitly open. Any future upgrade to strict-core selector closure or a directed physical sign datum must be exported and proven as a separate theorem-level object.

16 Operational physics proxies from projective closure

17 Mass-spectrum proxy (P694)

Let $H_\psi \in \mathbb{R}^{12\times 12}$ be the exported diagonal/local Psi-sector Hessian value instantiation (`F459`). We use its symmetrization:

$$H_\psi^{(s)} := \tfrac{1}{2}\big(H_\psi + H_\psi^\top\big).$$

For each Fourier pair plane pair_m ($m = 1, \ldots, 5$), the exported global projective selector closure (`F672`) provides a ray $u_m \in \mathbb{R}^{12}$ (defined only up to sign). Define the quadratic proxy:

$$m_m^2 := \frac{u_m^\top H_\psi^{(s)} u_m}{u_m^\top u_m}.$$

This is invariant under the sign gauge $u_m \mapsto -u_m$.

Probe `P694` computes these "mass-spectrum proxy" values on the current repo state (no Standard Model identification claimed):

pair	m^2 proxy (Rayleigh on $H_\psi^{(s)}$)
pair3	0.6687166508753918
pair5	1.3892705031823167
pair4	1.4474315384116325
pair2	2.0699508733172283
pair1	2.9337556210543343

18 Selector-aligned channel spectrum proxy (P696)

Probe `P696` refines `P694` into a 12-channel spectrum proxy by constructing a deterministic selector-aligned basis and inspecting $H_\psi^{(s)}$ in that basis.

On $\mathbb{Z}_{12}$, define the standard real orthonormal Fourier vectors:

$$e_0 := \frac{1}{\sqrt{12}}(1, \dots, 1), \qquad e_6 := \frac{1}{\sqrt{12}}(1, -1, 1, -1, \dots),$$

and for each $m = 1, \dots, 5$ the real pair-plane basis (c_m, s_m) with scale $\sqrt{2/12}$.

For each pair$_m$, `F672` exports coordinates (u_c, u_s) such that (in the declared Fourier convention)

$$u_m = u_c\, c_m + u_s\, s_m, \qquad u_m^\perp = (-u_s)\, c_m + (u_c)\, s_m,$$

so that $u_m^\perp$ is a deterministic orthogonal complement ray in the same plane (used only at ray/projector level).

Define the selector-aligned 12-channel basis:

$$B := \big(e_0,\ u_1,\ u_1^\perp,\ u_2,\ u_2^\perp,\ \dots,\ u_5,\ u_5^\perp,\ e_6\big),$$

and the induced matrix:

$$M := B^\mathsf{T}\, H_\psi^{(s)}\, B.$$

The diagonal entries of M are recorded as a simple operational "channel proxy spectrum". Mixing is audited explicitly by inspecting the off-diagonal norm.

On the current repo state, `P696` exports the sorted diagonal channel proxies (no Standard Model identification claimed):

channel	m^2 proxy
pair3:u	0.6687166508753918
pair5:u	1.3892705031823165
pair4:u	1.4474315384116314
pair2:u	2.069950873317227
e6	2.41906833577034
pair1:u	2.9337556210543316
pair5:u_perp	3.5360177064927694
pair4:u_perp	3.675931516755229
pair2:u_perp	4.2984508516608315
e0	4.761250376916641
pair3:u_perp	4.930970379004997
pair1:u_perp	5.08050282436479

The mixing audit is part of the exported result: on the current repo state, `P696` reports an off-diagonal-to-diagonal Frobenius ratio of 1.191284779529568 and a maximum off-block coupling Frobenius norm of 3.4009571444250914, so *no diagonalization claim* is made.

19 Basis-invariant Hessian eigen-spectrum proxy (F704)

The most basis-stable quadratic proxy spectrum is the invariant eigenvalue set of the symmetric Hessian $H_\psi^{(s)}$ itself. If we change coordinates by an orthogonal basis transformation $R \in O(12)$, then

$$H_\psi^{(s)} \mapsto R\, H_\psi^{(s)}\, R^\mathsf{T},$$

so the eigenvalues $\lambda_i\big(H_\psi^{(s)}\big)$ are unchanged. This avoids any basis-dependent "channel" choices and cleanly separates the invariant spectrum from mixing/assignment questions (which remain open at the physics-interpretation layer).

`F704` exports the invariant eigenvalue spectrum on the current repo state (dimensionless m^2 proxies in strict normalization; no physical unit claim):

mode index	$\lambda_i(H_\psi^{(s)})$ (proxy m^2)
0	0.08813142775481989
1	0.11195105316362809
2	0.23796767585569786
3	0.7641952808835573
4	0.7733323123191046
5	1.2068935563568985
6	1.7060617210611557
7	1.8861137936815946
8	2.290146094544166
9	2.989559197836824
10	12.356024722005182
11	12.800940342343853

20 What is and is not claimed physically

- **Claimed (operational):** the repo exports a strict projective selector closure and an OS bundle sufficient to compute the proxy spectra `P694`/`P696` and downstream observer-limit artifacts in a projective (ray-level) gauge-safe scope.
- **Not claimed:** that any proxy number is already identified with a Standard Model mass or coupling.
- **Not claimed:** that the selector-aligned channel basis diagonalizes H_ψ; mixing is audited and explicitly non-negligible (`P696`).
- **Not claimed:** kernel-alone/global `QW-2191` discharge, directed sign-sensitive physical orientation in strict core, or ToE closure.
- **Host matching discipline:** any attempted identification of proxy values with external physics data is necessarily non-strict and must carry an explicit external dataset + mapping policy; see `P702` (channel proxy) and `P704` (eigen-spectrum proxy). The current repo freezes this non-strict layer by separating:
  - the external dataset (targets only): `external_data/sm_mass_targets_v1.json`,
  - the external versioned policy (assignment + allowed DOF + optional thresholds): `external_data/sm_ho`

  where the current policy explicitly allows *only* a single global scale and sets `pass_criteria=null` (diagnostic metrics only; no pass/FAIL claim).
- **Non-strict diagnostic (current):** under the simplest one-scale centered-log assignment policy against an explicit PDG2024 mass target dataset, `P702` computes RMSE($\log m^2$) $\approx$ 8.10 and $\max|\Delta m/m| \approx 7.90 \times 10^2$, while `P704` computes RMSE($\log m^2$) $\approx$ 7.23 and $\max|\Delta m/m| \approx 4.35 \times 10^2$, so *no* "Standard Model match" claim is made at this stage.

21 Emergent observer pipeline (projective scope)

22 Observer-limit readout from projective closure output (P697)

From the exported global projective closure output projector B_{out} in the (o_+, o_-) basis (`F672`), `P697` exports a minimal observer-limit readout consisting of gauge-safe invariants of the 2×2 projector. Write

$$B_{\text{out}} = \begin{pmatrix} p_+ & c \\ c & p_- \end{pmatrix}.$$

Define:

$$y_{\text{bias}} := \tfrac{1}{2}(p_+ - p_-), \qquad y_{\text{total}} := \tfrac{1}{2}(p_+ + p_-), \qquad z_{\text{commit}} := p_+, \qquad z_{\text{residual}} := p_-.$$

On the current repo state, `P697` reports $p_+ = 1$ and $p_- \approx 8.37 \times 10^{-31}$, so the output projector is numerically the pure o_+ ray, and the readout invariants are $(y_{\text{bias}}, y_{\text{total}}, z_{\text{commit}}, z_{\text{residual}}) \approx (1/2, 1/2, 1, 0)$.

23 Projective emergent-observer chain computability from closure output (P699/N699)

In the declared projective/gauge-safe downstream scope, we represent a ray by a rank-1 (possibly unnormalized) symmetric positive semidefinite matrix P (a projector candidate). For a linear map M, the induced pushforward is:

$$P \;\mapsto\; M\,P\,M^{\mathsf{T}}.$$

Let Q_{out} be the exported global projective selector-closure output projector in the (o_+, o_-) basis (‘F672‘).

Define the downstream operator chain by the already exported matrices (‘F92..F99‘):

$$C = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad G = I_2, \quad H = I_2, \quad J = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

$$K = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad M = (1), \quad N = (1).$$

Starting from Q_{out}, define the induced chain:

$$Y = CQ_{\text{out}}C^{\mathsf{T}},\ Z = LYL^{\mathsf{T}},\ W = GZG^{\mathsf{T}},\ X = HWH^{\mathsf{T}},\ U = JXJ^{\mathsf{T}},\ F = KUK^{\mathsf{T}},\ P = MFM^{\mathsf{T}},\ C_{\text{cl}} = NPN^{\mathsf{T}}.$$

‘P699‘ computes this chain on the current repo state and audits that the declared residual components vanish (within tolerance) while commitment remains positive. ‘N699‘ packages the computability statement with explicit hard limits (no kernel-alone/global ‘QW-2191‘ discharge; no ToE closure; no actual emergent observer closure).

Theorem 23.1 (Projective emergent-observer chain computability (N699))**.** *On the current repo state, the repo exports a strict projective downstream chain computability witness: starting from the global projective selector closure output projector (`F672`), the induced projector pushforward semantics*

$$P \mapsto MPM^{\mathsf{T}}$$

along the exported operator chain (`F92..F99`) produces a fully computable chain of projectors up to the declared closure-candidate level (`P699`). This is a projective/ray-level (gauge-safe) computability result only.

Hard limits: *no kernel-alone/global `QW-2191` discharge, no ToE closure, and no actual emergent observer closure are claimed.*

24 Operational OS support packet (v1) (F698/P698/N698)

The v1 OS support packet (`F698`, audited by `P698`, packaged by `N698`) collects the minimum strict projective bundle sufficient to compute:

- global projective selector closure on C_{v1} (`N680`/`F672`),
- directed-vs-projective output-ray invariance and sign-lift gauge covariance (`N692`, `N693`),
- first quadratic proxy spectra (`P694`, `P696`),
- observer-limit readout (`P697`).

The mass-proxy meaning is frozen separately as a scope-limited theorem (`N703`). A basis-invariant Hessian eigen-spectrum mass observable is exported separately as `F704` and included in the v3 OS support packet (`F705`).

25 Operational OS support packet (v2) (F700/P700/N700)

The strict projective operational ToE "OS" support packet is exported in v2 form:

$$\Lambda_{\mathrm{OS}}^{(\mathrm{proj})} = \big(\texttt{N680},\ \texttt{N692},\ \texttt{N693},\ \texttt{P694},\ \texttt{P696},\ \texttt{P697},\ \texttt{N699}\big),$$

as the object `Lambda_strict_projective_operational_toe_os_support_v2` (`F700`), audited by `P700` and packaged as a theorem by `N700`. All hard limits remain explicit: no kernel-alone/global `QW-2191` discharge, no directed sign-sensitive physical orientation claim in strict core, and no ToE closure.

26 Operational OS support packet (v3) (F705/P705/N705)

The strict projective operational ToE "OS" support packet is exported in v3 form by explicitly including the basis-invariant Hessian eigen-spectrum mass observable `F704`:

$$\Lambda_{\mathrm{OS}}^{(\mathrm{proj})}(v3) = \big(\texttt{N680},\ \texttt{N692},\ \texttt{N693},\ \texttt{P694},\ \texttt{P696},\ \texttt{F704},\ \texttt{P697},\ \texttt{N699}\big),$$

as the object `Lambda_strict_projective_operational_toe_os_support_v3` (`F705`), audited by `P705` and packaged as a theorem by `N705`. All hard limits remain unchanged.

Theorem 26.1 (Strict projective operational OS support v3 (N705))**.** *On the current repo state, the repo exports one strict projective operational OS support packet v3:*

$$\Lambda_{\mathrm{OS}}^{(\mathrm{proj})}(v3) = \big(\texttt{N680},\ \texttt{N692},\ \texttt{N693},\ \texttt{P694},\ \texttt{P696},\ \texttt{F704},\ \texttt{P697},\ \texttt{N699}\big),$$

exported as `F705`, audited by `P705`, and packaged at theorem level by `N705`. This is an operational packaging result only: it adds an explicit tracked basis-invariant mass observable (`F704`) without strengthening any hard-limit claims.

Hard limits: *no kernel-alone/global `QW-2191` discharge, no directed physical sign datum, no Standard Model host matching, no ToE closure, and no actual emergent observer closure are claimed.*

Proposition 26.2 (Release-7 strict projective OS closure dashboard (P706))**.** *On the current repo state, the repo exports a single consolidated dashboard probe (`P706`) that checks the strict release-7 operational closure bundle by verifying: `N705` (OS support v3), `N701` (Operational ToE closure in strict projective OS scope), `N703` (mass-proxy meaning discipline), and `F704` (basis-invariant mass observable export). The current `P706` summary status is `PASS_RELEASE_7_STRICT_PROJECTIVE_OPERATIONAL_OS_CLOSURE_DASH`*

27 Open target: actual emergent observer closure (future)

The current strict projective pipeline proves *computability* of a downstream observer-facing chain, not the existence/uniqueness of an actual emergent observer. Any future "actual observer closure" claim must be exported as a separate theorem-level artifact with explicit premises and hard limits.

28 Hard limits and open problems

29 Kernel-alone/global QW-2191 discharge

`QW-2191` remains open at kernel-alone/global strict-core level. All global closure statements used in the OS bundle are in *projective* scope (rays/projectors) and do not assert strict-core uniqueness of directed selector representatives.

30 Directed physical sign datum

Directed/vector-level sign and orientation data are tracked as a gauge/convention layer. No strict-core theorem in Release 7 upgrades directed sign into a physical datum "for free".

31 Host matching and Standard Model identification

Any comparison of proxy spectra against external experimental data is non-strict and must carry:

- an explicit external dataset, and
- an explicit identification/mapping policy.

The repo exports such a harness (`P702`) and keeps it outside strict scope; current one-scale metrics do not support any "Standard Model match" claim.

32 What Release 7 declares as closed vs open

Closed (scope-limited, strict projective OS):

- projective global selector closure on C_{v1} with chart-independent output projector (`N680`/`F672`),
- operational OS support packets (v1–v3) and OS computability pipeline from closure output through observer-limit readout and downstream chain (`N698`, `N700`, `N705`, `N701`),
- strict basis-invariant Hessian eigen-spectrum mass observable (`F704`),
- residual sign $u_m \mapsto -u_m$ is gauge-irrelevant for the Release-7 OS observables actually used downstream (audited by `P709`, packaged by `N706`),
- a disciplined internal meaning of the quadratic "mass proxy" layer (`N703`).
- release-7 OS closure dashboard readiness (`P706`) (probe-level).

Open (explicit hard limits):

- kernel-alone/global `QW-2191` discharge,
- directed sign-sensitive physical orientation datum,
- Standard Model host matching as a strict claim,
- actual emergent observer closure.

33 Next steps (roadmap, no false pass)

This section records the next concrete moves under strict scientific discipline, separating strict targets from explicitly non-strict policy work.

Priority 1 (strict): `T173` — residual sign / kernel-alone `QW-2191` frontier

The strict frontier remains: (i) kernel-alone/global discharge of `QW-2191`, and (ii) any directed/sign-sensitive physical orientation datum in strict core beyond projective/ray semantics. Release 7 exports convention-layer directed representatives (`T174`/`T175`), but these do not upgrade sign into strict physics.

The next honest strict move under `T173` is therefore one of:

(a) export a genuinely strict internal selector source/ingredient that determines the residual sign in a chart-independent way (no hidden convention slots), or

(b) explicitly freeze the directed sign as a gauge/convention layer and keep physics claims restricted to projective/basis-invariant observables (operational closure already exists in that scope via `N701`).

For option (b), Release 7 now exports an explicit machine audit: `P709` enumerates all 2^5 residual sign patterns on $\{\text{pair1}, \ldots, \text{pair5}\}$ and verifies that the concrete OS numeric outputs used downstream (quadratic proxy layer `P694`/`P696` and the basis-invariant mass observable `F704`) are unchanged (within tolerance). The corresponding theorem-level package is `N706`.

The older methodology is also now audited directly rather than guessed at. The new audit `P711` (packaged by `N707`) shows the strongest honest reading: the `sigma_int`/topological lane survives as a genuine strict local ingredient class, reaching bridge discharge in declared `R1` scope, but it still does not provide the missing *global* chartwise sign-coherence provider on the full `C_v1` atlas. So it is a clue and a reusable ingredient, not yet a full `T173` discharge.

Priority 2 (explicitly non-strict policy): physical units (proxy → GeV)

Any unit assignment and any comparison to external experimental data must remain non-strict unless a strict unit map is derived. The repo already exports the required dataset-vs-policy separation:

- dataset: `external_data/sm_mass_targets_v1.json` (units = GeV),
- policy: `external_data/sm_host_matching_policy_v1.json` (one global scale; `pass_criteria=null`).

Additionally, Release 7 exports a dedicated non-strict *unit calibration* policy and a corresponding calibration-map probe for the basis-invariant proxy spectrum `F704`:

- policy: `external_data/proxy_to_gev_calibration_policy_v1.json` (`pass_criteria=null`),
- calibration map probe: `P710` (`fundamental_action_reconstruction/p710_current_nonstrict_proxy_to_`

The v1 calibration uses a single global scale parameter defined by matching geometric means of m^2 (assignment-free), and exports only a proxy→GeV interface (no identification/match claim).

Priority 3 (theory extension): "solitons = particles"

To make the slogan scientific, the program must export an explicit nonlinear EOM/Lagrangian in which soliton solutions are defined, a stability criterion, and a map from soliton moduli to the observer-limit proxy layer.

Priority 4 (falsifiable outputs): operational tests under frozen artifacts

The nearest falsifiable layer consistent with current strict scope is the basis-invariant proxy spectra (P694, P696, F704) and the non-strict external host-matching harnesses (P702, P704) producing diagnostics only (no pass claim).

A Export map: key artifacts

B Strict-core selector closure

- Global atlas + transition objects on C_{v1}: F469 (N515); exports SelectorAtlas_global_C_v1_strict_v1 and SelectorTransition_global_C_v1_strict_v1.
- Global projective selector state on C_{v1}: F470 (N516).
- Global projective selector closure output on C_{v1}: F672 (N680).
- Directed/convention-layer lifts and sign-fixing: F690..F693 (N692, N693), with explicit boundaries (no QW-2191 discharge).

C Operational OS support

- Proxy spectra: P694, P696; basis-invariant eigen-spectrum mass observable: F704; observer-limit readout: P697.
- Downstream computability chain: P699 packaged by N699.
- OS support packets: F698/P698/N698 (v1), F700/P700/N700 (v2), F705/P705/N705 (v3).
- Operational ToE closure statement: N701.
- Quadratic mass-proxy meaning definition: N703.
- Residual-sign gauge irrelevance audit (Release-7 OS scope): P709, packaged by N706.
- Previous-methodology survival/global-gap audit for T173: P711, packaged by N707.
- Non-strict host matching harnesses (external-data dependent): P702, P704.
- Release-7 strict projective OS closure dashboard: P706.

D Emergent observer artifacts

- Observer-limit readout invariants from the projective closure output projector: P697.
- Full induced operator chain computability under $P \mapsto MPM^{\mathsf{T}}$: P699/N699.

E Explicit numeric artifacts (Release 7)

F Frozen kernel distance profile on $\mathbb{Z}_{12}$

The strict kernel-mode lane uses the frozen symmetric mixing matrix K_{total} on $\mathbb{Z}_{12}$ (specialization route R14, instantiated in F459). On the current repo state, its cyclic-distance profile (for $d = 1, \ldots, 6$) is:

d	$k(d)$
1	0.46998567264502017
2	0.19204355169010282
3	0.09142861427792497
4	0.04702916874565040
5	0.02413122336363006
6	0.011070817321442113

The full 12×12 matrix is circulant/symmetric under cyclic distance:

$$(K_{\text{total}})_{ij} = k\big(d_{\text{cyc}}(i,j)\big), \qquad k(0) = 0, \qquad k(d) = k(12-d) \quad (d = 1, \ldots, 6).$$

G Explicit diagonal/local Psi-sector Hessian instantiation (F459)

The diagonal/local Psi-sector Hessian instantiation `F459` exports a symmetric matrix $H_\psi \in \mathbb{R}^{12\times 12}$. On the current repo state it decomposes *exactly* as:

$$(H_\psi)_{ij} = \begin{cases} m_0^2 + d_{\text{local}}(i), & i = j, \\ k\big(d_{\text{cyc}}(i,j)\big), & i \neq j, \end{cases}$$

with

$$m_0^2 = 1.013551972358388.$$

The exported diagonal residual profile $d_{\text{local}}(i)$ and the resulting diagonal are:

i	$d_{\text{local}}(i)$	$(H_\psi)_{ii} = m_0^2 + d_{\text{local}}(i)$
0	0.34269049977246646	1.3562424721308544
1	11.518361441687219	12.531913414045606
2	0.5284499241054783	1.5420018964638662
3	−0.1567291917971062	0.8568227805612817
4	−0.5341686256176337	0.4793833467407542
5	1.2939217196807558	2.3074736920391437
6	−0.5936675263840503	0.4198844459743376
7	1.2939217196807553	2.3074736920391430
8	−0.5341686256176337	0.4793833467407542
9	−0.1567291917971062	0.8568227805612817
10	0.5284499241054781	1.5420018964638660
11	11.518361441687219	12.531913414045606

All full numeric matrices (K_{total}, H_ψ) and the invariant eigen-spectrum are exported in `fundamental_action_re`

H Frozen non-strict host matching layer (dataset vs policy)

I External Standard-Model mass dataset (v1)

The repo freezes the external dataset (targets-only) as: `fundamental_action_reconstruction/external_data/sm` with dataset id `SM_mass_targets_v1_pdg2024_summary_tables`. For reader convenience, the current target list (units: GeV) is:

id	m (GeV)	σ_m (GeV)
e	0.00051099895	1.5×10^{-13}
u	0.00216	7.0×10^{-5}
d	0.0047	7.0×10^{-5}
μ	0.1056583755	2.3×10^{-9}
s	0.0935	8.0×10^{-4}
τ	1.77693	9.0×10^{-5}
c	1.273	4.6×10^{-3}
b	4.183	7.0×10^{-3}
W	80.3692	1.33×10^{-2}
Z	91.188	2.0×10^{-3}
H	125.2	1.1×10^{-1}
t	172.57	2.9×10^{-1}

J External host matching policy (v1, frozen no-pass)

The repo freezes the external matching policy as: `fundamental_action_reconstruction/external_data/sm_host` with policy id `SM_host_matching_policy_v1_centered_log_assignment_with_global_scale`. The policy declares:

- an assignment algorithm: linear sum assignment on squared centered $\log m^2$ differences,
- **allowed DOF:** only a single global scale factor (defined by centered $\log m^2$ matching after the assignment),
- **disallowed DOF:** sector scales, kinetic-metric refits, nonlinear mass maps,
- **pass criteria:** `pass_criteria=null`, with semantics: *no pass/FAIL claim is permitted* under this policy version (diagnostic metrics only).

Hard limits: this entire layer remains non-strict and external-data dependent. It is not part of the strict OS closure bundle.